

Reader: This limit will characterize the rate of change of the function f directly at $x=x_0$.

Author: Exactly. Let us calculate the limit in detail:

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta f(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

(1) and examine first of all the mathematical nature of

this limit.

Reader: Since point x_0 is fixed, it is evidently the limit of the ratio of two functions of Δx for $\Delta x \rightarrow 0$.

Author: Let us denote these functions by F and G :

$$F(\Delta x) = f(x_0 + \Delta x) - f(x_0), \quad G(\Delta x) = \Delta x$$

Reader: Limit (1) is then $\lim_{\Delta x \rightarrow 0} \frac{F(\Delta x)}{G(\Delta x)}$, where $\lim_{\Delta x \rightarrow 0} F(\Delta x) = 0$ and $\lim_{\Delta x \rightarrow 0} G(\Delta x) = 0$.

Hence, we have here a limit similar to that discussed at the end of Dialogue Seven, namely, a limit of the type $\frac{0}{0}$.

Author: Right. This limit, that is, the limit of the type $\frac{0}{0}$ is the main subject of this dialogue.

The primary requirement in this case is the existence of the limit. It means that the function f should be such that $\lim_{\Delta x \rightarrow 0} F(\Delta x) = 0$.

The necessary condition for satisfying this equality is the continuity of f at $x=x_0$. But we shall discuss this problem later.

If the limit of the type $\frac{0}{0}$ (in other words, limit (1)) does exist, it is called "the derivative of the function f at point $x=x_0$ " and usually denoted by $f'(x_0)$.